

EXCURSION REPORT

COPERNICUS HUBS AND INSTITUTIONS

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A visit to
Earth Observation Data
Center (EODC)
and
Umweltbundesamt (UBA)-
Environment Agency Austria



Submitted by :

Rabina Twayana

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On June 10, 2025, a one-day academic excursion to the Earth Observation Data Center (EODC) and the Environment Agency Austria (Umweltbundesamt (UBA)) was conducted, organized as part of the Copernicus Hubs and Institutions course by the GeoInformatics Department, Paris Lodron University Salzburg.

EARTH OBSERVATION DATA CENTRE (EODC), VIENNA

Time and Duration: 11:00 AM- 2:00 PM

Agenda: The introductory presentation by **Christian Briese**, CEO of the Earth Observation Data Centre for Water Resources Monitoring GmbH, followed by guided tours to the different rooms where the large-scale hardware infrastructure that supports Earth Observation (EO) data storage, processing, and distribution was observed.

About EODC

EODC was established on May 15, 2014, as a Public-Private Partnership (PPP). Its primary aim is to foster collaboration between the public and private sectors as well as international partners, particularly through the establishment and operation of shared IT infrastructure. Additionally, EODC seeks to serve as a bridge between scientific research and practical applications, promoting the effective use of Earth Observation (EO) data across various domains. This structure underpins EODC's robust capabilities, including storage and access to a petabyte-scale repository of Copernicus Sentinel data, advanced cloud and HPC infrastructure, specialized EO processing tools, and a strong cooperative network of experts from academia, government, and industry.

Currently, they have the archive for Sentinel 1, Sentinel 2, and Sentinel 3 EO datasets, ECMWF Reanalysis v5 (ERA-5), and Wegener Center (WEGC) Global Navigation Satellite System (GNSS) radio occultation (RO) record. EODC is involved in several Copernicus initiatives such as Copernicus Global Land Service (C-GLOPS), Copernicus Climate Change Service (C3S), Global Flood Monitoring Service (GFM), etc.

Insights from EODC

The introductory presentation by Christian provides valuable insights about the center's mission, available facilities, and offered services. He also shared overviews of their leading ongoing projects/activities such as FAIR2earth, ScaleFloodS, and Interface (INformation accEss Service for Austrian CopErnicus). It was a great opportunity to witness how the Petabytes of EO dataset are systematically stored and backed up. In addition to this, observing and understanding the architecture, design, operation, and maintenance of high-performance computing servers such as Vienna Scientific Cluster (VSC-4 & VSC-5) as shown in Figure 1 and 2 below, was very interesting. Additionally, one of the most intriguing parts was knowing about the liquid (water) based cooling technology to ensure optimal performance, reliability, and longevity of high-performance computing (HPC) systems. The server rooms were quieter than expected because of this liquid cooling technology. One of the key highlights for me was learning about the MUSICA (Multi-Site Computer Austria) project, which is a collaborative initiative involving seven Austrian universities, funded by the Austrian Research Promotion Agency (FFG) and the Austrian Ministry of Education, Science and Research. Their goal is to

transform the existing high-performance computing infrastructure federated high-performance computing system across multiple sites, including Vienna, Linz, and Innsbruck, and support for HPDA/ML/AI and compatibility with the European HPC ecosystem.

Additionally, another interesting thing was to witness the automated data handling infrastructure in rape room within which petabytes of Earth observation (EO) data are managed and stored. The facility uses robotic tape libraries where robotic systems automatically move magnetic tape cartridges in and out of drives based on user requests. This approach provides high-capacity, energy-efficient data management, reduced human intervention, and secure long-term data archiving.

Overall, it was amazing to witness how high-performance computing and Earth observation come together to support climate research, disaster monitoring, environmental protection, and global sustainability.

Observation at EODC

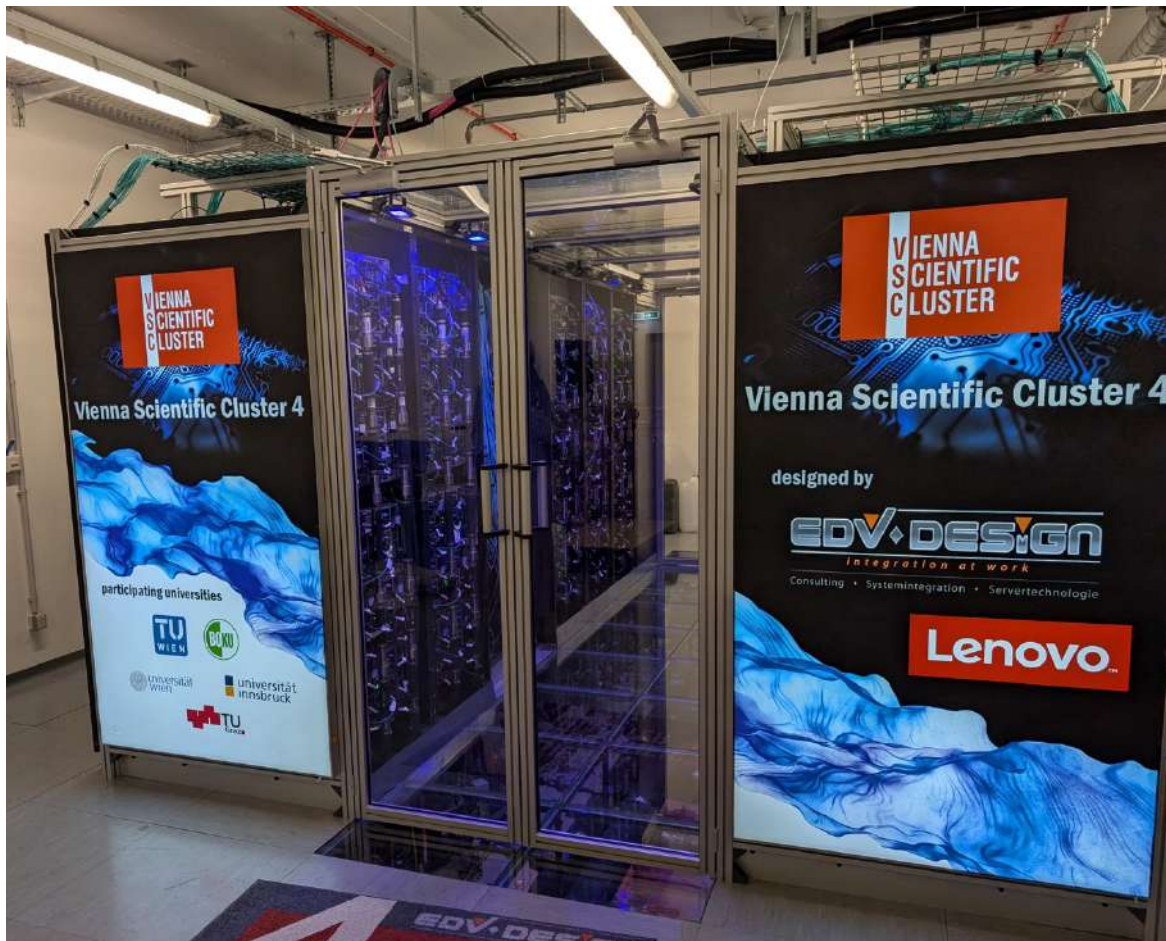


Figure 1: Vienna Scientific Cluster 4



Figure 2: Vienna Scientific Cluster 5



Figure 3: MUSICA Quantum Austria



Figure 4: Automated robotic storage system



Figure 5: Water-based Cooling System

ENVIRONMENT AGENCY AUSTRIA (UMWELTBUNDESAMT (UBA))

Time and Duration: 14:30 - 16:30 PM

Location: Spittelauer Lände 5, 1090 Vienna, Austria

Agenda: The presentation by Gebhard Banko (team leader, Remote Sensing and Analysis) and Roland Grillmayer (senior expert, GIS and Remote Sensing). The main purpose was to know about the different projects led by the team to support environmental monitoring, climate change assessment, land use mapping, and policy development.

About UBA

The Umweltbundesamt is Austria's important environmental expert organization. It was created by law in 1985 and became a limited liability company in 1999. Over the past 40 years, the agency has been playing a key role in the transformation of the economy and society to ensure sustainable living. Their experts provide the basis for decision-making at local, regional, and international levels. This organization is also internationally active, providing environmental consulting and supporting institutional capacity building in more than 60 countries. Interdisciplinary work is central to the organization, with experts specializing in a wide range of scientific fields, including energy & environmental management, industrial environmental protection, environmental engineering & water management, agricultural economics, business administration, environmental technology, forestry, project management, and many others. It employs around 600 experts as of 2023, excluding contract-based external international experts and project staff, out of which approximately 50% are women staff. They are dedicated to transparency and impartiality, engaging with stakeholders across various sectors such as politics, administration, business, science, and civil society to provide decision-making support, identify risks and opportunities, and develop sustainable solutions, and offer advice on forward-thinking approaches to meet the challenges at the environment/economy/society interface.

Insights from UBA

The overall presentation given by Gebhard and Roland provides a comprehensive overview of the projects that support environmental monitoring. One of the major projects discussed was the operational monitoring of land take and soil sealing in Austria. The team combines administrative GIS data (forest, buildings, OSM, spatial planning zonation) with high-resolution satellite imagery to produce detailed, up-to-date maps of settlement expansion, infrastructure and energy development, and soil sealing, which are critical for sustainable spatial planning and climate adaptation.

Another important teamwork was on drought vegetation monitoring, which is relevant given Austria's increasing drought risk. In this project, they combine the optical and microwave remote sensing along with in-situ, laser scanning, to derive vegetation indices and assess the vegetation dynamics across various spatial and temporal scales, and develop the indicator "Drought Impact on Ecosystem". They also compare the Copernicus products with the global and national datasets to evaluate their suitability to support the development of EO-based indicators proposed by the "Forest Monitoring Regulations"

A significant part of the learning experience was seeing how the Umweltbundesamt contributes to capacity building and knowledge transfer through the EU4Environment and EU4GreenRecoveryEast programme on Earth Observation, Copernicus, environmental indicators to support land cover mapping, water accounting activities, and develop environmental reporting. The team's leadership in developing geodata infrastructures, semantic web and AI technologies, and digital twins for environmental data management illustrated the

importance of interoperability and innovation in the digital transformation of environmental policy. I enjoyed the live demonstration of a digital twin based on a knowledge graph that is built on top of Neo4J for complex time series analysis by referencing geo-objects (geo-data).

In a nutshell, the presentation offered deep insights on the application of remote sensing, GIS, and AI in monitoring land use, biodiversity, water quality, and climate impacts.



Figure 6: Presentation by Gebhard Banko [left], Roland Grillmayer [right]

Summary on EU Inspire and Copernicus Data Space Ecosystem

INSPIRE (Infrastructure for Spatial Information in the European Community) was launched in 2007 under the Directive (2007/2/EC) of the European Parliament. It is a European Union initiative and operated by EU member states, aimed at the sharing of environmental spatial data and information among public sector organizations and better facilitating public access to spatial information across Europe. The users who get benefits from geoinformation available through INSPIRE include governmental agencies and public administration, utility and public services, commercial and professional users, research and educational institutions, non-governmental and non-profit organizations, as well as individual citizens. It aims at creating a standardized, interoperable spatial data infrastructure across member states to address major challenges associated with interoperability of geoinformation among member countries of European Union that includes variety of data formats, different spatial reference systems, differences in data models, differences in data quality, variety of portrayal rules, differences in metadata standards and many more. This is because of a lack of coordination regarding geographic information strategies across borders and between governments. With this standardization, it has been possible to combine the spatial information from different sources across Europe seamlessly, regardless of language and national boundaries. In a nutshell, harmonized data, services, and metadata are key pillars of interoperability. The INSPIRE Directive has 34 spatial data themes needed for environmental applications, as well as provides technical requirements for their publication via network services. The INSPIRE geoportal provides the central access point into the INSPIRE metadata, data, and services.

Copernicus Data Space Ecosystem (CDSE) is the EU's platform that provides access to all historical data available from the Sentinel satellite constellations. The CDSE is led by leading European cloud and earth observation service providers. With the help of European Space Agency (ESA) guidance and expertise from the partners, CDSE delivers a comprehensive, high-quality, and user-friendly ecosystem. In addition to data access, it also provides data processing capabilities within the system, which make it a go-to platform for anyone looking to extract insights from Copernicus data. It supports users in accessing, viewing, using, downloading, and analyzing data. It also allows user to integrate their datasets and deploy applications at various levels of federation. The main purpose of CDSE is to empower users with the tools and resources they need to unlock the full potential of this data. The service aims to support users in building various applications needed to provide timely, accurate, and support for environmental monitoring, climate change, land use, emergency response, and many more. The Ecosystem Services Registry showcases a diverse range of services designed to support users at every stage of their workflow. It enables users and providers to discover, contribute, and scale Earth observation solutions, turning Copernicus data into real-world applications and operational value.

Web Links

EODC: <https://eodc.eu/>

UBA: <https://www.umweltbundesamt.at/>

INSPIRE: <https://data.jrc.ec.europa.eu/collection/id-0109>

CDSE: <https://dataspace.copernicus.eu/about>